Math Kangaroo Practice

Grade 3-4 · 40 questions · harder · seed 78400

Part A: Each question is worth 3 points

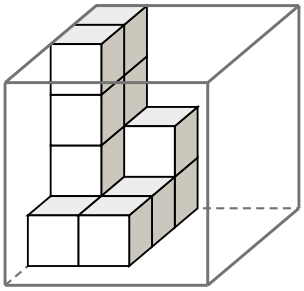
1. Mia writes all the numbers from 989 to 1014.
How many digits does Mia write?

A. 75 ***B.*** 78 ***C.*** 93 ***D.*** 94 ***E.*** 104

2. The odometer of a car shows 536948 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 31 ***B.*** 23 ***C.*** 22 ***D.*** 21 ***E.*** 12

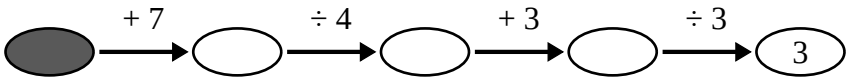
3. Nina has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Nina has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



At most how many more such cubes can Nina put into the aquarium?

A. 50 ***B.*** 51 ***C.*** 52 ***D.*** 54 ***E.*** 64

4. Emma wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 3.

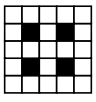


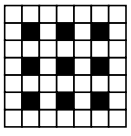
What is Emma's favourite number?

A. 3 ***B.*** 16 ***C.*** 17 ***D.*** 18 ***E.*** 21

5. These diagrams follow a pattern of black and white cells. If the pattern continues, how many white cells will the next diagram have?

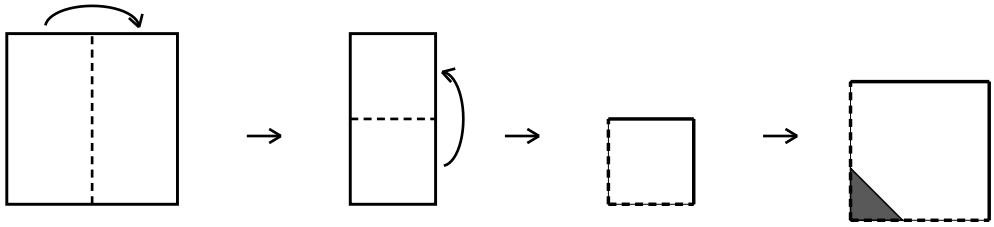
8 white cells


21 white cells

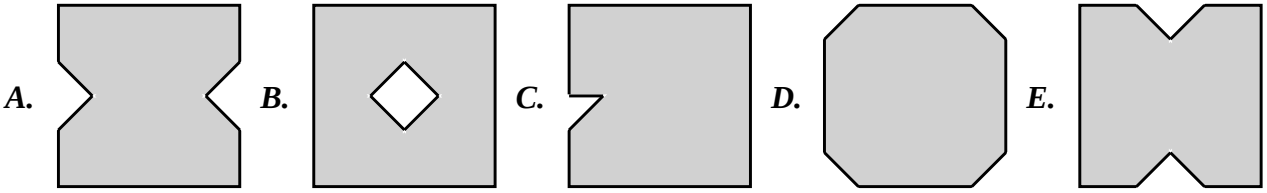

40 white cells

A. 81 ***B.*** 65 ***C.*** 64 ***D.*** 61 ***E.*** 59

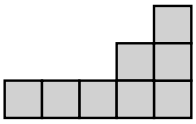
6. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). The dark part is cut off, then the paper is unfolded.



What does the paper look like?



7. Tom wants to add one more piece to the grey piece to make the smallest possible rectangle.



How many little squares must that piece have?

A. 10 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

8. Five ribbons have these lengths.
Which ribbon is the longest?

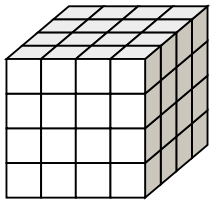
A. 800 cm ***B.*** 8600 mm ***C.*** 79 dm ***D.*** 8 m 10 cm ***E.*** 85 dm

9. The digits from 1 to 9 are written on 9 cards. Fred has 3, 4 and 6; Alex has 1, 2 and 8; Nina has 5, 7 and 9. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who **cannot** get 27 as the result?

A. Fred ***B.*** Alex ***C.*** Nina
D. All can get 27 ***E.*** Alex and Nina

Part B: Each question is worth 4 points

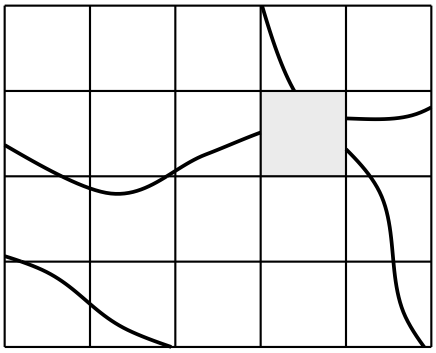
10. A cube made of 4 × 4 × 4 small cubes stands on the table and is painted grey on all sides you can reach: the top and the four sides, but **not** the bottom. Then it is taken apart.



How many small cubes have exactly 3 painted faces?

A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

11. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).



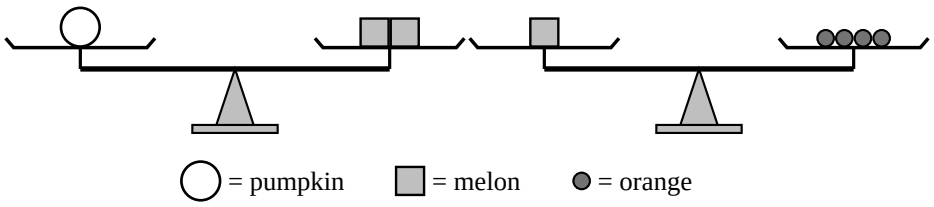
Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

12. Chloe, Fabienne, Ravi, Hana and Tom each have a different pet: a rabbit, a hamster, a parrot, a cat and a dog. Fabienne doesn't like hamsters. Ravi has a pet that barks. Fabienne doesn't like cats. Hana has a pet with two legs. Chloe doesn't like hamsters.
Which statement is true?

A. Fabienne has a rabbit ***B.*** Tom has a rabbit ***C.*** Hana has a dog
D. Ravi has a hamster ***E.*** Tom has a cat

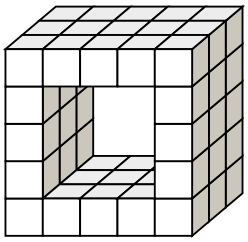
13. The two balances are level. One pumpkin weighs as much as 2 melons, and one melon weighs as much as 4 oranges.



How many oranges weigh as much as 2 pumpkins?

A. 16 ***B.*** 17 ***C.*** 18 ***D.*** 19 ***E.*** 20

14. Mia built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Mia wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

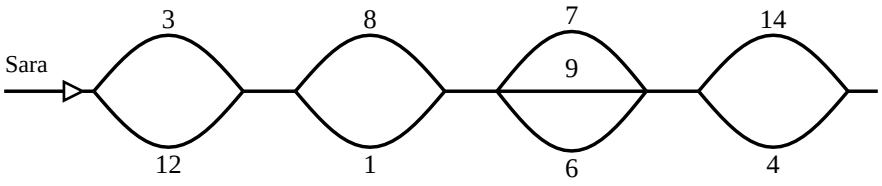
A. 57 ***B.*** 27 ***C.*** 21 ***D.*** 18 ***E.*** 9

15. Fred has the cards 4, 5, 6 and 9. Fred must use every card exactly once, with +, –, ×, ÷ and brackets.

Which number can Fred **not** get?

A. 66 ***B.*** 65 ***C.*** 62 ***D.*** 59 ***E.*** 57

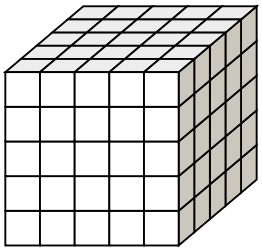
16. Sara walks along the road from left to right and adds up the numbers along the way.



What is the biggest total Sara can get?

A. 64 ***B.*** 49 ***C.*** 48 ***D.*** 43 ***E.*** 14

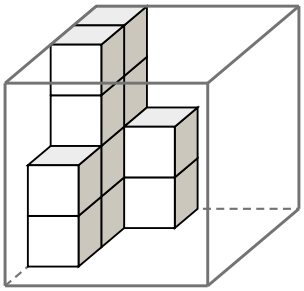
17. Leo glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have paint on exactly two faces?

A. 54 ***B.*** 45 ***C.*** 42 ***D.*** 40 ***E.*** 36

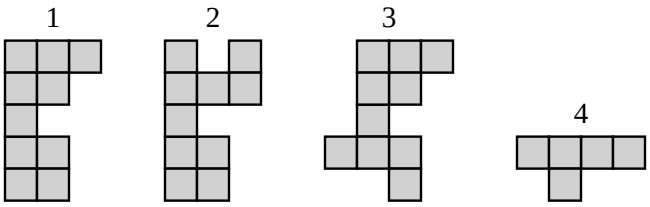
18. Nina puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Now Nina wants to cover the whole bottom of the box with one layer of cubes.



How many more cubes are needed for that?

A. 52 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 4

19. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.

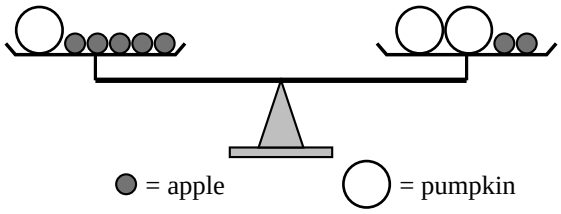


A. 2 and 4 ***B.*** 1 and 3 ***C.*** 3 and 4 ***D.*** 1 and 4 ***E.*** 2 and 3

20. Mia threads 29 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 2 more blue beads than big red beads.
How many small red beads are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

21. The balance is level: 1 pumpkin and 5 apples on one plate, 2 pumpkins and 2 apples on the other.



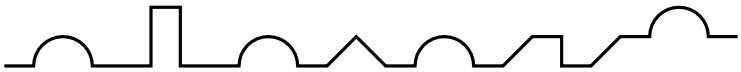
One pumpkin weighs the same as how many apples?

A. 1 apple ***B.*** 2 apples ***C.*** 3 apples ***D.*** 6 apples ***E.*** 7 apples

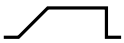
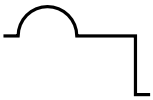
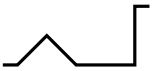
22. Eva has a rope 4 m long and cuts it into pieces that are each 40 cm long.
How many pieces does Eva get?

A. 10 ***B.*** 9 ***C.*** 8 ***D.*** 7 ***E.*** 1

23. This is the outline of a town seen from the river at sunset.



Which piece **can** be part of the outline?

A. ***B.***  ***C.*** ***D.***  ***E.*** 

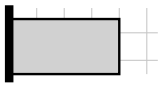
24. What is the smallest number whose digits add up to 26?

A. 998 ***B.*** 989 ***C.*** 902 ***D.*** 900 ***E.*** 899

25. The seats in a theatre are numbered from 1 to 112.
How many times is the digit 5 written?

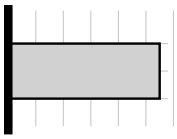
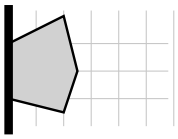
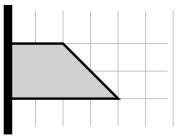
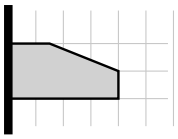
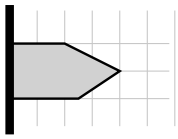
A. 21 ***B.*** 22 ***C.*** 23 ***D.*** 24 ***E.*** 26

26. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments. Spread out flat, the flag is 4 squares wide and 2 squares tall (see the picture).



the flag, flat

Which shadow **cannot** be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

27. Nina thinks of a number, takes away 7, then multiplies by 3, then multiplies by 2, and gets 6. Which number did Nina think of?

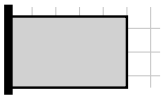
A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 10

Part C: Each question is worth 5 points

28. Exactly 303 digits are used to write all the house numbers in a street, starting from 1. How many houses are there?

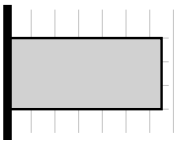
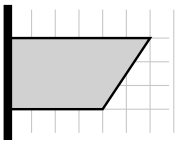
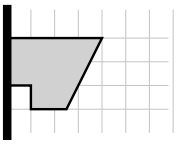
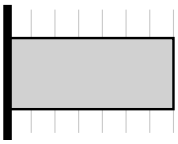
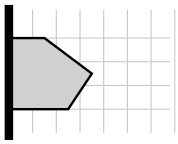
A. 146 ***B.*** 137 ***C.*** 136 ***D.*** 47 ***E.*** 38

29. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Karla draws the shadow at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Four of the drawings are wrong.



the flag, flat

Which drawing can be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

30. How many three-digit numbers have digits that add up to 4?

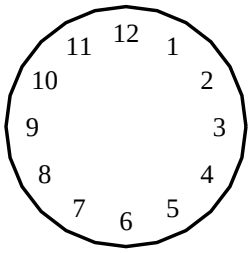
A. 20 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 10

31. A bar code is made of 18 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 2 more white bars than wide black bars.

How many narrow black bars are there?

A. 4 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

32. The face of a clock is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 2 and 3, and between 8 and 9

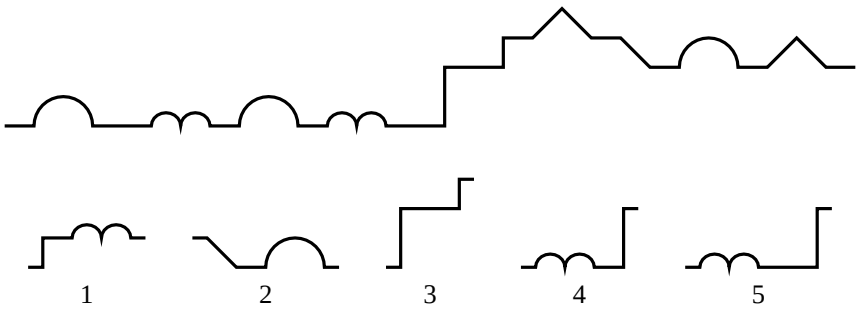
B. between 4 and 5, and between 8 and 9

C. between 11 and 12, and between 4 and 5

D. between 3 and 4, and between 9 and 10

E. between 4 and 5, and between 7 and 8

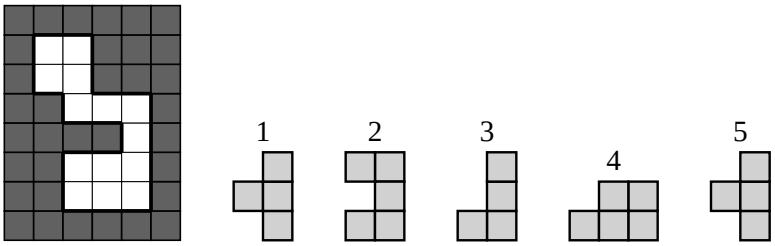
33. Far away we see the skyline of a castle.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

34. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+3+4 ***C.*** 2+3+4 ***D.*** 2+4+5 ***E.*** 3+4+5

35. Basil and Peter are twins. Basil was born at 11:54 p.m. on December 31, 2009. Peter was born 19 minutes later.
What is Peter's date of birth?

A. January 1, 2010 ***B.*** December 31, 2010 ***C.*** January 2, 2010
D. January 1, 2008 ***E.*** December 31, 2009

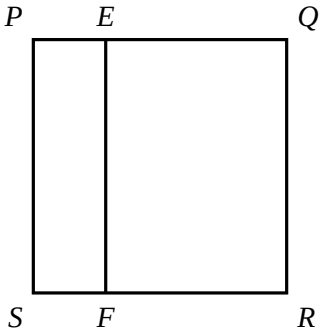
36. Hugo paints a flag with three stripes: top, middle and bottom. Hugo has 5 colours (red, green, blue, orange and yellow). All three stripes must be different colours.
How many different flags can Hugo paint?

A. 10 ***B.*** 12 ***C.*** 15 ***D.*** 20 ***E.*** 60

37. How many different letters are used to write the two words BUTTERFLY and COMPUTER?

A. 12 ***B.*** 13 ***C.*** 14 ***D.*** 16 ***E.*** 17

38. The square *PQRS* is cut into two rectangles. The perimeter of *PEFS* is 18 cm and the perimeter of *EQRF* is 24 cm.



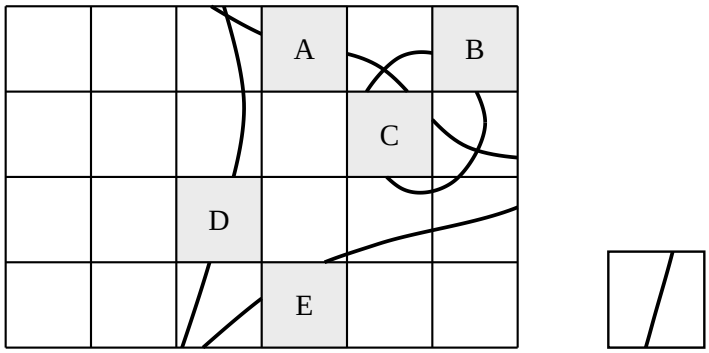
How long is the side of the square *PQRS*?

A. 7 cm ***B.*** 6 cm ***C.*** 5 cm ***D.*** 4 cm ***E.*** 3 cm

39. Ravi, Noah, Chloe, Julien and Omar each have a different pet: a hamster, a cat, a goldfish, a turtle and a canary. Omar has a pet that can fly. Noah doesn't like cats. Noah owns a four-legged creature. Julien has a pet with a shell. Chloe owns a four-legged creature.
Which statement is **not** true?

A. Omar has a canary ***B.*** Noah has a hamster ***C.*** Omar has a goldfish
D. Chloe has a cat ***E.*** Ravi has a goldfish

40. Some paths go across a park. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

Answer Key

1.	**C**	**2.**	**C**	**3.**	**B**	**4.**	**C**	**5.**	**B**	**6.**	**B**
7.	**B**	**8.**	**B**	**9.**	**E**	**10.**	**E**	**11.**	**B**	**12.**	**A**
13.	**A**	**14.**	**B**	**15.**	**E**	**16.**	**D**	**17.**	**E**	**18.**	**C**
19.	**D**	**20.**	**D**	**21.**	**C**	**22.**	**A**	**23.**	**A**	**24.**	**E**
25.	**A**	**26.**	**A**	**27.**	**C**	**28.**	**B**	**29.**	**E**	**30.**	**E**
31.	**C**	**32.**	**D**	**33.**	**C**	**34.**	**C**	**35.**	**A**	**36.**	**E**
37.	**A**	**38.**	**A**	**39.**	**C**	**40.**	**D**				

Solutions

1. C (93) — There are 26 numbers: 11 × 3 + 15 × 4 = 93 digits. [2007 Q23 · across_hundred]

2. C (22) — Counting up, every number before 536970 repeats a digit. 536970 has all different digits: 536970 – 536948 = 22. [2002 Q24 · all_different]

3. B (51) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 13 inside (count column by column, also the hidden ones). 64 – 13 = 51. [2007 Q14 · how_many_more]

4. C (17) — Go backwards and do the opposite: 3 × 3 = 9, 9 – 3 = 6, 6 × 4 = 24, 24 – 7 = 17. [2007 Q5 · favourite_number]

5. B (65) — The sides grow by 2 and the black cells are 1 × 1, 2 × 2, 3 × 3, ... The next diagram is 9 × 9 = 81 cells with 4 × 4 = 16 black: 81 – 16 = 65. [2007 Q21 · next_white]

6. B — The cut corner is the centre of the paper (look where the folds are). Mirrored four times it gives a hole in the middle: design B. [2007 Q24 · which_result]

7. B (7) — The smallest rectangle around the grey piece is 5 × 3 = 15 squares. 15 – 8 = 7. [2007 Q8 · squares_missing]

8. B (8600 mm) — In cm: 800 cm = 800 cm, 8600 mm = 860 cm, 79 dm = 790 cm, 8 m 10 cm = 810 cm, 85 dm = 850 cm. The longest is 8600 mm. [2003 Q10 · longest]

9. E (Alex and Nina) — Fred: 3 + 4 × 6 = 27. Alex: with 1, 2 and 8 nothing gives 27 (the closest are 24 = (1 + 2) × 8); Nina: with 5, 7 and 9 nothing gives 27 (the closest are 26 = 5 × 7 – 9 and 28 = (9 – 5) × 7). [2002 Q21 · who_cannot]

10. E (4) — Only the 4 top corners: they have the top and two sides painted. The bottom corners lose their bottom face. [2007 Q17 · not_underneath]

11. B — Look where the lines of the picture reach the grey gap: top: 1 line, right: 2 lines, bottom: no line, left: 1 line. Only square B has its lines in exactly those places in this position. The others are mirrored, mirrored and turned, upside down. [2002 Q1 · turned_square]

12. A (Fabienne has a rabbit) — The only possible matching: Chloe – cat; Fabienne – rabbit; Ravi – dog; Hana – parrot; Tom – hamster. [2002 Q18 · which_true]

13. A (16) — One pumpkin = 2 melons = 2 × 4 = 8 oranges; 2 pumpkins = 2 × 8 = 16 oranges. [2002 Q9 · chain]

14. B (27) — The hole is 3 × 3 = 9 cubes in the front and goes 3 cubes back: 9 × 3 = 27. [2002 Q20 · fill_the_hole]

15. E (57) — (4 + 6) × 5 + 9 = 59; 4 × (5 + 9) + 6 = 62; 4 × (6 + 9) + 5 = 65; 4 × 9 + 5 × 6 = 66. But with 4, 5, 6 and 9 nothing gives 57 (the closest are 55 = 5 + 6 × 9 – 4 and 59 = (4 + 6) × 5 + 9). [2002 Q21 · which_cannot]

16. D (43) — Take the bigger number at every fork: 12 + 8 + 9 + 14 = 43. [2007 Q1 · largest_sum]

17. E (36) — They are on the edges but not at the corners: 12 × 3 = 36. [2003 Q9 · painted]

18. C (12) — The bottom has 4 × 4 = 16 places. 4 of them already have a column of cubes. 16 – 4 = 12. [2007 Q14 · bottom_layer]

19. D (1 and 4) — Pieces 1 and 4 have 15 squares together and fit into a 5 × 3 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

20. D (3) — There are 15 red and 14 blue beads. Big red: 14 – 2 = 12. Small red: 15 – 12 = 3. [2003 Q17 · necklace]

21. C (3 apples) — Take 1 pumpkin and 2 apples off both plates. Left: 1 pumpkin = 3 apples. [2002 Q9 · both_sides]

22. A (10) — 4 m = 400 cm. 400 ÷ 40 = 10. [2003 Q10 · pieces]

23. A — Only this piece is found in the line. The others don't fit — B: its flat part between the two features has the wrong length; C: it has two small bumps where the line has a pointed roof; D: it steps down where the line steps up; E: it has a pointed roof where the line has a round dome. [2002 Q7 · can_belong]

24. E (899) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 899: 8 + 9 + 9 = 26. [2007 Q4 · smallest_with_sum]

25. A (21) — As the last digit: 11 times; as the tens digit: 10 times. Total 21. [2003 Q16 · count_one_digit]

26. A — Turning or folding can only make the flag look narrower. Picture A reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

27. C (8) — Go backwards: 6 ÷ 2 = 3, 3 ÷ 3 = 1, 1 + 7 = 8. [2007 Q5 · think_of_a_number]

28. B (137) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 303 – 189 = 114 digits are three-digit numbers: 114 ÷ 3 = 38, from 100 to 137. [2003 Q16 · three_digit_pages]

29. E — Check two things: the pole edge must be covered all the way, and nothing may reach more than 5 squares from the pole. Only picture E passes both checks. [2002 Q15 · only_one_right]

30. E (10) — Count by the first digit: 4 + 3 + 2 + 1 = 10. [2007 Q4 · count_with_sum]

31. C (2) — Starting black and ending white, the bars come in pairs: 9 black and 9 white. Wide: 9 – 2 = 7. Narrow: 9 – 7 = 2. [2003 Q17 · ends_differ]

32. D (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [2002 Q11 · straight_crack]

33. C (3) — Pieces 2, 3, 5 fit. Not: 1 (it has two small bumps where the line has a pointed roof); 4 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

34. C (2+3+4) — The empty area has 14 squares, so the three sizes must add up to 14. Only 2 + 3 + 4 also fit together in the right shape. [2003 Q18 · three_pieces]

35. A (January 1, 2010) — 6 minutes after 11:54 p.m. it is midnight, and a new day and a new year start. Peter was born 13 minutes after midnight, on January 1, 2010. So Basil is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]

36. E (60) — Top stripe: 5 choices, middle: 4, bottom: 3. 5 × 4 × 3 = 60. [2002 Q22 · stripes]

37. A (12) — BUTTERFLY uses 8 different letters and COMPUTER uses 8. E, R, T and U are in both, so count them once: 8 + 8 – 4 = 12. [2007 Q3 · letter_cards]

38. A (7 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 18 + 24 = 42 cm = 6 sides. Side = 42 ÷ 6 = 7 cm. [2002 Q6 · two_perimeters]

39. C (Omar has a goldfish) — The only possible matching: Ravi – goldfish; Noah – hamster; Chloe – cat; Julien – turtle; Omar – canary. [2002 Q18 · which_not_true]

40. D (gap D) — The square has top: 1 line, right: no line, bottom: 1 line, left: no line. Only gap D has lines reaching it at exactly those places. [2002 Q1 · which_gap]